Ruchi Parekar

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Summary

Engineering student with hands-on experience in IoT, embedded systems, and PCB design. Proficient in using Proteus for circuit simulation and testing. Strong in problem-solving, hardware troubleshooting, and creating reliable electronic solutions. Enthusiastic learner who adapts quickly and works effectively in collaborative projects.

Education

- **JSPM's Rajarshi Shahu College of Engineering, Tathawde** 2023-27
  Bachelor of Technology in Electronics and Telecommunication Engineering CGPA: 9.17
- **SMT. S. S. Ajmera High School Junior College, Pune, India** Passing: 2023
  Board – SSC, Science – Physics, Chemistry, and Maths 75.50%
- **St. Alphonsa High School** Passing: 2021
  Board – SSC 85.40%

Experience

- **Project Intern- Sunay Enterprises, Pune** July 2025

  This project was motivated by the need to reduce liquid wastage and eliminate manual monitoring in domestic and industrial tanks. The goal was to develop an IoT-based liquid level monitoring system capable of providing continuous level measurement and real-time alerts. The system successfully delivered accurate monitoring and timely notifications, improving operational efficiency, safety, and minimizing human intervention.

Projects

- **Smart Fire Detection System Using ESP32** Dec 2025-Present
  - Identified the need for early fire detection and remote alerts to reduce fire-related risks and damage.
  - To design an IoT-based fire detection system using ESP32, sensors, and Blynk for real-time monitoring and alerts.
  - Successfully achieved timely fire detection with instant mobile notifications, improving safety and rapid response.

- **ESP32 Wi-Fi & Hotspot Based LED Control System** Oct 2025-Nov 2025
  - Identified the need for flexible wireless device control both with and without external Wi-Fi connectivity.
  - To enable LED control using ESP32 through a mobile phone via IP address in both Wi-Fi station mode and self-created hotspot (Access Point) mode.
  - Successfully achieved real-time LED control using a smartphone over external Wi-Fi and ESP32 hotspot, demonstrating standalone and network-based IoT control.

- **Obstacle Detection Robot** Aug 2025-Oct 2025
  - Identified the need to prevent liquid overflow, wastage, and manual monitoring in domestic and industrial tanks.
  - To design an obstacle detection robot using sensors and a microcontroller for autonomous obstacle avoidance.
  - Successfully achieved reliable obstacle detection and avoidance, improving navigation accuracy and operational safety.

- **IoT-Based Smart Liquid Level Monitoring System** June 2025-July 2025
  - Identified the need to prevent liquid overflow, wastage, and manual monitoring in domestic and industrial tanks.
  - To design a smart IoT-based system for continuous liquid level monitoring with real-time alerts and automation.
  - Achieved accurate level detection and instant alerts, improving efficiency, safety, and reducing human intervention.name for this project.

- **PIC18 Microcontroller-Based Calculator (Proteus Simulation)** March 2025-May 2025
  - Identified the need to understand microcontroller interfacing, logic implementation, and simulation-based testing
  - To design and simulate a basic calculator using a PIC18 microcontroller in Proteus for arithmetic operations.
  - Successfully implemented and verified correct calculator operations through Proteus simulation, strengthening skills in PIC programming and embedded system design.

- **Minute Chronometric Systems** Oct 2024-Dec 2024
  - Identified the need for a simple, accurate, and low-cost minute-based time tracking system for learning and timing applications.
  - To design and implement a minute chronometric system using 555 Timer, 7490 counters, and 7447 BCD-to-7-segment drivers.
  - Successfully achieved precise and stable minute counting, validated through Proteus simulation and hardware testing, suitable for educational and industrial timer applications.

SKILLS

- **Soft Skills :** Skilled in clear and persuasive communication, analytical problem-solving, cooperative teamwork, maintaining composure under pressure, delivering client-focused solutions, and ensuring precision in every task.
- **Technical Skills :** Intermediate C/C++, Python, Embedded C
- **Tools :** MATLAB, Arduino IDE, STM Cube IDE, Proteus, Simulink (Basic), MPLAB IDE

POSITION AND RESPONSIBILITY

Training and Placement Coordinator | E&TC Department
Student Advisor | IEI Student Chapter
Vice President | ENTESA